SSC MTS 2024 question paper with answers

8 October 2024 · Shift 2

79 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 11 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 14 questions

Q.2

In a mixture of 60 litres, milk and water are in the ratio 7 : 3. If 3 litres of water be added, find the ratio of milk and water in the new mixture.

(A) 3:2
(B) 1:2
(C) 2:1
(D) 2:3

Q.3

Find the simple interest (in ₹) on ₹3000 as a sum borrowed at 8% per year rate of interest for 5 years.

(A) 1240
(B) 1200
(C) 1160
(D) 1140

Q.5

Three red lights on the same road start blinking at the same time and blink repeatedly after every 4, 6 and 8 minutes, respectively. How many times will they blink together in an interval of 3 hours, excluding blinking together at the starting moment?

(A) 5
(B) 7
(C) 4
(D) 6

Q.6

Find the simple interest (in ₹) on ₹2000 as a sum borrowed at 7% per year rate of interest for 4 years.

(A) 500
(B) 560
(C) 600
(D) 520

Q.7

A man sold an article for ₹247.50, thereby gaining 12.5%. The cost of the article (in ₹) was:

(A) 210
(B) 220
(C) 224
(D) 225

Q.8

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 70 kg and 35 kg is:

(A) 51
(B) 52
(C) 49
(D) 50

Q.10

By selling an article for ₹20, a dealer makes a profit of 10%. By how much should he increase his selling price (in ₹, rounded off to two decimal place) so as to make a profit of 40%?

(A) 5.45
(B) 6.45
(C) 8.45
(D) 7.45

Q.11

The average of the present ages of P and Q is 56 years and their ages are in the ratio 5:3. What will be the age of Q seven years hence?

(A) 52 years
(B) 42 years
(C) 54 years
(D) 49 years

Q.12

The ratio of income to expenditure of a family is 18:15. Find the savings if the income of the family is ₹81,000.

(A) ₹15,000
(B) ₹17,500
(C) ₹12,750
(D) ₹13,500

Q.14

A car travels from Kanpur to Delhi at an average speed of 60 km/h and returns at an average speed of 90 km/h. Find the average speed of the car in the whole journey.

(A) 65 km/h
(B) 72 km/h
(C) 78 km/h
(D) 60 km/h

Q.15

The average weight of three friends Ram, Ramesh and Shyam is 76 kg. Somesh joins them and the average weight of the four friends becomes 74 kg. If Mohan, whose weight is 2 kg more than that of Somesh replaces Ramesh, the average weight of Ram, Shyam, Somesh and Mohan becomes 68 kg. Find the weight of Ramesh.

(A) 84 kg
(B) 89 kg
(C) 94 kg
(D) 75 kg

Q.16

A tunnel elevator can carry a maximum of 15 passengers with an average weight of 81 kg. However, four persons beyond the maximum capacity entered the elevator, resulting in an average weight of 82 kg and causing it to be overloaded. Find the average weight (in kg) of the additional four persons.

(A) 85.75
(B) 90.41
(C) 102.37
(D) 100

Q.17

In a 5-km race, A beats B by 750 metres and C by 1260 metres. By how many metres does B beat C in the same race?

(A) 700 metres
(B) 600 metres
(C) 400 metres
(D) 500 metres

Q.20

The curved surface area of a cylinder is three-fifth of its total surface area. Find the ratio between the height and radius of the cylinder.

(A) 8 : 3
(B) 3 : 5
(C) 2 : 5
(D) 3 : 2

Reasoning Ability and Problem Solving 16 questions

Q.1

DFIE is related to CGHF in a certain way based on the English alphabetical order. In the same way, MOFC is related to LPED. To which of the following is GRRP related, following the same logic?

(A) ERQQ
(B) FSQQ
(C) FSQR
(D) DSPQ

Q.2

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: No teacher is a male. No male is an Engineer. Conclusions: I. No teacher is an Engineer. II. Some male are teachers.

(A) Only conclusion II follows.
(B) Neither conclusion I nor II follows.
(C) Only conclusion I follows.
(D) Both conclusions I and II follow.

Q.3

In a certain code language, 'CHOSEN' is coded as ‘YTMIWN’ and 'COMMON' is coded as ‘YMOOMN’. How will 'CLIENT' be coded in the same language?

(A) YPTXNG
(B) YPSWNH
(C) YPSXNG
(D) YPSVNF

Q.5

8 is related to 93 following a certain logic. Following the same logic, 10 is related to 117. To which of the following is 14 related, following the same logic? (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 189
(B) 165
(C) 166
(D) 187

Q.8

In a certain code language, ‘did we know’ is written as ‘kp as th’ and ‘we solve this’ is written as ‘nk kp bo’. How is ‘we’ written in the given language?

(A) as
(B) bo
(C) kp
(D) nk

Q.9

What should come in place of the question mark (?) in the given series based on the English alphabetical order? EBF, IGG, MLH, QQI ,?

(A) TUJ
(B) UVJ
(C) SUI
(D) SUJ

Q.10

In a certain code language, ‘EACH’ is coded as ‘6138’ and ‘CAME’ is coded as ‘1396’. What is the code for ‘M’ in that language?

(A) 1
(B) 9
(C) 3
(D) 6

Q.11

The second number in the given number pairs is obtained by performing certain mathematical operation(s) on the first number. Select the pair in which the numbers are related in the same way as are the numbers of the given pairs. (NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into their constituent digits. E.g. 13 – Operations on 13 such as adding /deleting /multiplying to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.) 6, 42; 9, 63

(A) 13, 91
(B) 11, 67
(C) 17, 102
(D) 19, 123

Q.12

What should come in place of the question mark (?) in the given series? 39, 47, 55, 63, 71, ?

(A) 80
(B) 78
(C) 79
(D) 77

Q.14

Seven people P, U, L, M, A, G and E are sitting around a circular table, facing the centre (but not necessarily in the same order). P is sitting third to the left of M. G is sitting second to the right of M. Only one person is sitting between M and A when counted from the left of M. L is sitting to the immediate right of G. Only one person is sitting between L and E . Who is sitting to the immediate left of U?

(A) A
(B) G
(C) M
(D) P

Q.15

In a certain code language, 'toy is new' is coded as 'bz tq gl' and 'give me toy' is coded as 'hx vm tq'. How is 'toy' coded in the given language?

(A) hx
(B) vm
(C) bz
(D) tq

Q.16

T, U, V, W, X, and Y are sitting around a circular table facing the centre (but not necessarily in the same order). U is sitting third to the right of V, who is sitting second to the left of W. Y is not an immediate neighbour of U. T is sitting to the immediate left of V. Who is sitting to the immediate left of T?

(A) U
(B) X
(C) V
(D) W

Q.17

CA 19 is related to ID 14 in a certain way. In the same way, ML 53 is related to SO 48. To which of the following is GD 8 related, following the same logic?

(A) NF –3
(B) NH 3
(C) MG 3
(D) MH –3

Q.18

Read the given statement(s) and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statement(s). Statements: Some cups are glasses. No cup is a plate. Conclusions: (I) No glass is a plate. (II) Some glasses are not cups.

(A) Only conclusion I follows.
(B) Neither conclusion I nor II follows.
(C) Only conclusion II follows.
(D) Both conclusions I and II follow.

Q.19

What will come in the place of '?' in the following equation, if '÷' and '–' are interchanged and '×' and '+' are interchanged? 12 × 96 – 24 + 8 ÷ 38 = ?

(A) 6
(B) 16
(C) –14
(D) –4

Q.20

What will come in the place of '?' in the following equation, if '+' and '–' are interchanged and '×' and '÷' are interchanged? 56 – 33 × 3 ÷ 4 + 36 =?

(A) 68
(B) 64
(C) 74
(D) 66

General Awareness 25 questions

Q.1

The National Food Security Act (NFSA) was enacted in which year by the Government of India?

(A) 2008
(B) 2005
(C) 2013
(D) 2001

Q.2

Prerana Shrimali received the Sangeet Natak Akademi Award in which of the following years for her contribution to the Kathak dance form?

(A) 2015
(B) 2003
(C) 2012
(D) 2009

Q.3

According to the Ministry of Agriculture and Farmers Welfare 2022-23, which state of India is the largest producer of sugarcane?

(A) Andhra Pradesh
(B) Karnataka
(C) Bihar
(D) Uttar Pradesh

Q.4

India won its first Olympic gold in hockey in which of the following years?

(A) 1936
(B) 1928
(C) 1932
(D) 1940

Q.5

As per HDI 2021, which of the following neighbouring countries of India had the highest female life expectancy at birth?

(A) Maldives
(B) Sri Lanka
(C) Bangladesh
(D) Bhutan

Q.6

From 13 September 2023, DMK MP Tiruchi Siva has been appointed the chairperson of the Committee on _____.

(A) Information Technology
(B) Industry
(C) Home
(D) Transport

Q.7

Which of the following Rights is mentioned in Article 21 of the Indian Constitution?

(A) Right against Exploitation: Prohibition of trafficking in human beings and forced labour

(B) Right to Freedom of Religion: Freedom to manage religious affairs

(C) Right to Freedom: Protection of life and personal liberty

(D) Right to Equality: Equality before law

Q.8

How much money was allotted by the Gujarat government in August 2023 for the One District, One Product (ODOP) Programme to promote its indigenous products at the international level?

(A) ₹58 crore

(B) ₹10 crore

(C) ₹100 crore

(D) ₹25 crore

Q.9

The Sahitya Akademi award winning novel 'Zindaginama' was written by:

(A) Amrita Pritam

(B) Krishna Sobti

(C) Mahadevi Verma

(D) Mannu Bhandari

Q.10

In which among the following years was Hindu widow remarriage declared legal in India?

(A) 1899

(B) 1856

(C) 1872

(D) 1893

Q.11

Gabbro is an example of which type of rocks?

(A) Metamorphic rock

(B) Organically formed sedimentary rock

(C) Sedimentary rock

(D) Igneous rock

Q.12

The remains of Great Bath have been found from the excavation at which of the following Harappan sites?

(A) Harappa

(B) Mohenjodaro

(C) Lothal

(D) Kalibangan

Q.13

Which of the following peaks is NOT located in the Himalayan mountains?

(A) Nanda Devi

(B) Kalsubai

(C) Annapurna

(D) Kamet

Q.14

Annapurna Devi is associated with which of the following musical instruments?

(A) Flute

(B) Mridangam

(C) Surbahar

(D) Sarangi

Q.15

What is the length of the men's kabaddi court?

(A) 13 m

(B) 17 m

(C) 11 m

(D) 15 m

Q.16

In the context of microfinance in India, the concept of SHG-BLP is used. What does L stand for in SHG-BLP?

(A) Linkage

(B) Lending

(C) Lend

(D) Link

Q.17
Who regarded Vedas as the source of all knowledge and known for the slogan ‘Go back to the Vedas’?

(A) Swami Vivekananda
(B) Raja Rammohun Roy
(C) Keshav Chandra sen
(D) Swami Dayanada Saraswati

Q.18
What is the retirement age of Judges serving at the Supreme Court of India?

(A) 65 years
(B) 62 years
(C) 60 years
(D) 68 years

Q.19
Rigveda is the earliest specimen of:

(A) Indo European language
(B) Indo-African language
(C) Indian language
(D) Russian language

Q.20
The famous ‘Tripartite struggle’ for supremacy occurred between which of the following powers?

(A) Gurjar-Pratihara, Rashtrakuta and Palas
(B) Chalukya, Sen and Pratihars
(C) Sen, Pala and Rashtrakuta
(D) Chouhan, Pala and Gurjar

Q.21
Who among the following is worshipped as Nataraja before many classical dance performances in India?

(A) Indra
(B) Vishnu
(C) Brahma
(D) Shiva

Q.22
Chithirai festival of Tamil Nadu is dedicated to which of the following Goddesses?

(A) Mariamman
(B) Pidari
(C) Meenakshi
(D) Muneeswaran

Q.23
Which gas is released when you combine vinegar and baking soda in a balloon, causing it to inflate?

(A) Hydrogen
(B) Oxygen
(C) Carbon dioxide
(D) Nitrogen

Q.24
Four resistors of equal resistance R each are connected in various possible ways both in series and/or parallel to get different combined resistances. Which of the following values of resistance cannot be achieved by using all the four resistors?

(A) 1.33 R
(B) 0.25 R
(C) 2 R
(D) 2.5 R

Q.25
Which of the given options is rod-shaped, carries genetic information, is visible only at cell division and is present in the nucleus?

(A) Cytoplasm
(B) Lysosomes
(C) Chromosomes
(D) Ribosomes

English Language and Comprehension 24 questions

Q.1

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. They were / interested / on / learning new languages.

(A) interested
(B) learning new languages
(C) on
(D) They were

Q.2

Select the most appropriate option to substitute the underlined segment in the given sentence. The project “fell besides due to lack” of funding and support.

(A) fell through due to lack
(B) fell until due to lack
(C) fell about due to lack
(D) fell along due to lack

Q.3

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. Sarah / will singing / at the concert / on Saturday.

(A) at the concert
(B) Sarah
(C) will singing
(D) on Saturday

Q.4

Select the most appropriate option that can substitute the underlined segment in the given sentence. If there is no need to substitute it, select ‘No substitution required’. The White man had to swallow his bombast and “have completed the journey in mortified”.

(A) No substitution required
(B) complete the journey in mortification
(C) completes the journey mortifying
(D) completing the mortifying journey

Q.5

Select the most appropriate synonym of the following word. Reluctant

(A) Unrelenting
(B) Curious
(C) Pleasant
(D) Unwilling

Q.6

Select the most appropriate synonym of the underlined word. The management gave “kudos” to Jithin for his performance.

(A) punishment
(B) admonishment
(C) praise
(D) criticism

Q.7

Select the INCORRECTLY spelt word.

(A) Extraneous
(B) Quiescent
(C) Posthumous
(D) Colloquiel

Q.8

Select the most appropriate option to fill in the blank. The way she elaborated on every detail with precision impressed everyone _____ the meeting.

(A) at
(B) off
(C) above
(D) into

Q.9

The following sentence has been split into four segments. Identify the segment that contains a grammatical error. I hope that / yesterday's lunch / at Rahul's place / was at your liking.

(A) was at your liking.
(B) at Rahul's place
(C) I hope that
(D) yesterday's lunch

Q.11
Select the most appropriate option to fill in the blank. The new intern was eager to learn the _____ of the corporate world.

(A) rides
(B) flies
(C) ropes
(D) tides

Q.12
The following sentence has been split into four segments. Identify the segment that contains a grammatical error. While she / was watched / the movie, / her phone rang.

(A) her phone rang
(B) While she
(C) the movie,
(D) was watched

Q.13
Select the most appropriate synonym of the given word. Hamper

(A) Hinder
(B) Accuse
(C) Urge
(D) Absorb

Q.14
Select the most appropriate synonym of the given word. Panacea

(A) Remedy
(B) Pressure
(C) Survey
(D) Miracle

Q.15
Select the appropriate option that can substitute the underlined idiom in the given sentence. Everyone says that waking up early in the morning is beneficial for our health and overall well-being. But it is “easier said than done”.

(A) a step towards better health
(B) a life filled with excitement
(C) to start performing better than before
(D) not as easy as it seems to be

Q.16
Select the most appropriate verb / verb phrase to fill in the blank. In 2030, advancements in technology _____ the way we live, work and interact with each other.

(A) could be reshaped
(B) will be reshaping
(C) shall have been reshaped
(D) have reshaped

Q.17
Select the correct spelling of the underlined word in the given sentence. The flight was about to take off when suddenly the authorities came to ensure the record of the “maintanance” department.

(A) maintainence
(B) maintenance
(C) maintnance
(D) maintainance

Q.18
Select the most appropriate option to fill in the blank. During the meeting, the employer's _____ was clear as he calmly addressed each team member's concerns and provided insightful solutions.

(A) composes
(B) composed
(C) composure
(D) compose

Q.19
Select the most appropriate synonym of the underlined word in the given sentence. The “lenient” ways of the manager prompted substandard work from his subordinates.

(A) easygoing
(B) insane
(C) irritating
(D) ignorant

Q.20

Parts of the following sentence have been given as options. Select the option that contains an error. My sister-in-law is the most succeeded for all the industrialists in this country today.

(A) succeeded for all
(B) My sister-in-law is the most
(C) in this country today
(D) the industrialists

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A phobia is an irrational fear of an object, situation or living thing. While fear is (1)_____ natural response to danger, phobias often occur (2)_____ response to something unlikely to cause real harm. In some cases, a person with a (3)_____ will meet the diagnostic criteria for a specific phobia, which is a type of anxiety disorder. Someone with a specific phobia will (4)_____ intense fear and anxiety when confronted with the source of the phobia, which could include blood, heights or specific animals like dogs. Phobias are (5)_____ by significant distress and often cause a person to avoid the source of their fear or suffer from extreme anxiety when faced with it. Someone experiencing a phobia that is interfering with everyday life should consider undergoing treatment for their phobia, as outcomes for professional care are typically favourable.

Select the most appropriate option to fill in blank number 1.

(A) these
(B) those
(C) one
(D) a

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A phobia is an irrational fear of an object, situation or living thing. While fear is (1)_____ natural response to danger, phobias often occur (2)_____ response to something unlikely to cause real harm. In some cases, a person with a (3)_____ will meet the diagnostic criteria for a specific phobia, which is a type of anxiety disorder. Someone with a specific phobia will (4)_____ intense fear and anxiety when confronted with the source of the phobia, which could include blood, heights or specific animals like dogs. Phobias are (5)_____ by significant distress and often cause a person to avoid the source of their fear or suffer from extreme anxiety when faced with it. Someone experiencing a phobia that is interfering with everyday life should consider undergoing treatment for their phobia, as outcomes for professional care are typically favourable.

Select the most appropriate option to fill in blank number 2.

(A) of
(B) on
(C) at
(D) in

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A phobia is an irrational fear of an object, situation or living thing. While fear is (1)_____ natural response to danger, phobias often occur (2)_____ response to something unlikely to cause real harm. In some cases, a person with a (3)_____ will meet the diagnostic criteria for a specific phobia, which is a type of anxiety disorder. Someone with a specific phobia will (4)_____ intense fear and anxiety when confronted with the source of the phobia, which could include blood, heights or specific animals like dogs. Phobias are (5)_____ by significant distress and often cause a person to avoid the source of their fear or suffer from extreme anxiety when faced with it. Someone experiencing a phobia that is interfering with everyday life should consider undergoing treatment for their phobia, as outcomes for professional care are typically favourable.

Select the most appropriate option to fill in blank number 3.

(A) trouble
(B) danger
(C) phobia
(D) violence

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A phobia is an irrational fear of an object, situation or living thing. While fear is (1)_____ natural response to danger, phobias often occur (2)_____ response to something unlikely to cause real harm. In some cases, a person with a (3)_____ will meet the diagnostic criteria for a specific phobia, which is a type of anxiety disorder. Someone with a specific phobia will (4)_____ intense fear and anxiety when confronted with the source of the phobia, which could include blood, heights or specific animals like dogs. Phobias are (5)_____ by significant distress and often cause a person to avoid the source of their fear or suffer from extreme anxiety when faced with it. Someone experiencing a phobia that is interfering with everyday life should consider undergoing treatment for their phobia, as outcomes for professional care are typically favourable.

Select the most appropriate option to fill in blank number 4.

(A) dream
(B) experience
(C) aspire
(D) horizon

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. A phobia is an irrational fear of an object, situation or living thing. While fear is (1)_____ natural response to danger, phobias often occur (2)_____ response to something unlikely to cause real harm. In some cases, a person with a (3)_____ will meet the diagnostic criteria for a specific phobia, which is a type of anxiety disorder. Someone with a specific phobia will (4)_____ intense fear and anxiety when confronted with the source of the phobia, which could include blood, heights or specific animals like dogs. Phobias are (5)_____ by significant distress and often cause a person to avoid the source of their fear or suffer from extreme anxiety when faced with it. Someone experiencing a phobia that is interfering with everyday life should consider undergoing treatment for their phobia, as outcomes for professional care are typically favourable.

Select the most appropriate option to fill in blank number 5.

(A) qualified
(B) classified
(C) characterised
(D) categorised

Answer key

Numerical and Mathematical Ability

Q.2 **(C)**	Q.3 **(B)**	Q.5 **(B)**	Q.6 **(B)**	Q.7 **(B)**	Q.8 **(D)**
Q.10 **(A)**	Q.11 **(D)**	Q.12 **(D)**	Q.14 **(B)**	Q.15 **(C)**	Q.16 **(A)**
Q.17 **(B)**	Q.20 **(D)**				

Reasoning Ability and Problem Solving

Q.1 **(B)**	Q.2 **(B)**	Q.3 **(B)**	Q.5 **(B)**	Q.8 **(C)**	Q.9 **(B)**
Q.10 **(B)**	Q.11 **(A)**	Q.12 **(C)**	Q.14 **(A)**	Q.15 **(D)**	Q.16 **(B)**
Q.17 **(C)**	Q.18 **(B)**	Q.19 **(A)**	Q.20 **(B)**		

General Awareness

Q.1 **(C)**	Q.2 **(D)**	Q.3 **(D)**	Q.4 **(B)**	Q.5 **(A)**	Q.6 **(B)**
Q.7 **(C)**	Q.8 **(A)**	Q.9 **(B)**	Q.10 **(B)**	Q.11 **(D)**	Q.12 **(B)**
Q.13 **(B)**	Q.14 **(C)**	Q.15 **(A)**	Q.16 **(A)**	Q.17 **(D)**	Q.18 **(A)**
Q.19 **(A)**	Q.20 **(A)**	Q.21 **(D)**	Q.22 **(C)**	Q.23 **(C)**	Q.24 **(C)**
Q.25 **(C)**					

English Language and Comprehension

Q.1 **(C)**	Q.2 **(A)**	Q.3 **(C)**	Q.4 **(B)**	Q.5 **(D)**	Q.6 **(C)**
Q.7 **(D)**	Q.8 **(A)**	Q.9 **(A)**	Q.11 **(C)**	Q.12 **(D)**	Q.13 **(A)**
Q.14 **(A)**	Q.15 **(D)**	Q.16 **(B)**	Q.17 **(B)**	Q.18 **(C)**	Q.19 **(A)**
Q.20 **(A)**	Q.21 **(D)**	Q.22 **(D)**	Q.23 **(C)**	Q.24 **(B)**	Q.25 **(C)**

Explanations

Numerical and Mathematical Ability

Q.2 · Answer (C) 2:1

The mixture has 42 litres of milk and 18 of water. With 3 more litres of water it is 42 : 21 = 2 : 1.

Q.3 · Answer (B) 1200

SI = 3000 × 8 × 5/100 = ₹1200.

Q.5 · Answer (B) 7

They blink together every LCM(4, 6, 8) = 24 minutes. 180 ÷ 24 = 7.5, so 7 times in 3 hours after the start.

Q.6 · Answer (B) 560

SI = 2000 × 7 × 4/100 = ₹560.

Q.7 · Answer (B) 220

Cost price = 247.50 ÷ 1.125 = ₹220.

Q.8 · Answer (D) 50

(40 + 49 + 56 + 70 + 35) ÷ 5 = 250 ÷ 5 = 50 kg.

Q.10 · Answer (A) 5.45

Cost price = 20/1.1. For a 40% profit he must sell at (20/1.1) × 1.4 = 28/1.1 = ₹25.45, an increase of ₹5.45.

Q.11 · Answer (D) 49 years

Their ages add up to 2 × 56 = 112, so Q is 3/8 × 112 = 42 now and 49 in seven years.

Q.12 · Answer (D) ₹13,500

Savings are 18 − 15 = 3 of the 18 parts of income: 81,000 × 3/18 = ₹13,500.

Q.14 · Answer (B) 72 km/h

Over equal distances the average speed is 2 × 60 × 90/(60 + 90) = 10,800/150 = 72 km/h.

Q.15 · Answer (C) 94 kg

Ram + Ramesh + Shyam = 228 kg, and with Somesh 4 × 74 = 296, so Somesh is 68 kg and Mohan 70 kg. Ram + Shyam = 4 × 68 − 68 − 70 = 134, so Ramesh = 228 − 134 = 94 kg.

Q.16 · Answer (A) 85.75

15 × 81 = 1,215 kg and 19 × 82 = 1,558 kg, so the four extra persons weigh 343 kg: 343 ÷ 4 = 85.75 kg each on average.

Q.17 · Answer (B) 600 metres

When A finishes 5,000 m, B has run 4,250 m and C 3,740 m. While B runs 5,000 m, C runs 3,740 × 5,000/4,250 = 4,400 m, so B wins by 600 m.

Q.20 · Answer (D) 3 : 2

$2\pi rh = (3/5) \times 2\pi r(h + r)$ gives $5h = 3h + 3r$, so $2h = 3r$ and $h : r = 3 : 2$.

Reasoning Ability and Problem Solving

Q.1 · Answer (B) FSQQ

The letters move −1, +1, −1, +1: D → C, F → G, I → H, E → F and M → L, O → P, F → E, C → D. So G → F, R → S, R → Q, P → Q: FSQQ.

Q.2 · Answer (B) Neither conclusion I nor II follows.

Two negative statements give no definite conclusion about teachers and engineers, so I does not follow; and as no teacher is a male, II cannot follow.

Q.3 · Answer (B) YPSWNH

Each letter's place in the alphabet and its code's place add up to 28: C (3) → Y (25), H (8) → T (20), N (14) → N (14). So CLIENT → YPSWNH.

Q.5 · Answer (B) 165
The number is multiplied by 12 and 3 taken away: 8 × 12 − 3 = 93 and 10 × 12 − 3 = 117. So 14 × 12 − 3 = 165.

Q.8 · Answer (C) kp
'we' is the only word in both sentences and 'kp' the only code in both, so we = kp.

Q.9 · Answer (B) UVJ
First letters move +4 (E, I, M, Q, U), second letters +5 (B, G, L, Q, V) and third letters +1 (F, G, H, I, J): UVJ.

Q.10 · Answer (B) 9
E, A and C are in both words, as are 6, 1 and 3 in both codes. The letter left in CAME is M and the digit left in its code is 9, so M = 9.

Q.11 · Answer (A) 13, 91
The second number is 7 times the first: 6 × 7 = 42 and 9 × 7 = 63. Likewise 13 × 7 = 91.

Q.12 · Answer (C) 79
Each term is 8 more than the one before: 71 + 8 = 79.

Q.14 · Answer (A) A
Counting to the right from M: M, E, G, L, P, A, U. G is second to the right of M and L to G's immediate right; P is third and A second to the left of M; E, with one person between it and L, takes the seat between M and G, and U the one left over. A is to U's immediate left.

Q.15 · Answer (D) tq
'toy' is the only word in both sentences and 'tq' the only code in both, so toy = tq.

Q.16 · Answer (B) X
Counting to the right from V: V, Y, W, U, X, T. W is second and U third to the right of V, and T to V's immediate left; Y, not beside U, takes the seat between V and W, and X the one between U and T. X is to T's immediate left.

Q.17 · Answer (C) MG 3
First letters move +6 (C → I, M → S), second letters +3 (A → D, L → O) and the number falls by 5 (19 → 14, 53 → 48). So GD 8 → MG 3.

Q.18 · Answer (B) Neither conclusion I nor II follows.
The cups that are glasses are not plates, but other glasses may be, so I does not follow; and every glass may be a cup, so II does not follow either.

Q.19 · Answer (A) 6
After interchanging the signs: 12 + 96 ÷ 24 × 8 − 38 = 12 + 32 − 38 = 6.

Q.20 · Answer (B) 64
After interchanging the signs: 56 + 33 ÷ 3 × 4 − 36 = 56 + 44 − 36 = 64.

General Awareness

Q.1 · Answer (C) 2013
The National Food Security Act, 2013 entitles up to 75% of the rural and 50% of the urban population to subsidised foodgrains.

Q.2 · Answer (D) 2009
The Sangeet Natak Akademi gave Prerana Shrimali its award for Kathak for the year 2009.

Q.3 · Answer (D) Uttar Pradesh
Uttar Pradesh grows the most sugarcane in India, well ahead of Maharashtra and Karnataka.

Q.4 · Answer (B) 1928
India won its first Olympic hockey gold at Amsterdam in 1928, with Dhyan Chand its top scorer.

Q.5 · Answer (A) Maldives
SSC's key gives the Maldives, whose overall life expectancy at birth, 79.9 years, is the highest of these four countries in the UNDP's Human Development Report 2021-22.

Q.6 · Answer (B) Industry
When Parliament's standing committees were reconstituted with effect from 13 September 2023, Tiruchi Siva, a DMK member of the Rajya Sabha, was named chairperson of the Committee on Industry.

Q.7 · Answer (C) Right to Freedom: Protection of life and personal liberty
Article 21, protection of life and personal liberty, is one of the Right to Freedom articles (19 to 22).

Q.8 · Answer (A) ₹58 crore
Gujarat's ₹58 crore grant of August 2023 went to more than 25 handicraft products from 21 districts, among them Patan's Patola and Jamnagar's Bandhani.

Q.9 · Answer (B) Krishna Sobti
Krishna Sobti won the Sahitya Akademi Award in 1980 for her Hindi novel 'Zindaginama'.

Q.10 · Answer (B) 1856
The Hindu Widows' Remarriage Act, 1856, championed by Ishwar Chandra Vidyasagar, made the remarriage of Hindu widows legal.

Q.11 · Answer (D) Igneous rock
Gabbro is a coarse-grained intrusive igneous rock, formed as magma cools slowly deep underground.

Q.12 · Answer (B) Mohenjodaro
The Great Bath was found at Mohenjodaro, in Sindh.

Q.13 · Answer (B) Kalsubai
Kalsubai, the highest peak in Maharashtra, is in the Sahyadri (Western Ghats). Nanda Devi, Kamet and Annapurna are Himalayan peaks.

Q.14 · Answer (C) Surbahar
Annapurna Devi, daughter of Ustad Allauddin Khan, was a celebrated player of the surbahar, the bass sitar.

Q.15 · Answer (A) 13 m
A men's kabaddi court measures 13 m × 10 m; the women's is 12 m × 8 m.

Q.16 · Answer (A) Linkage
SHG-BLP is the Self Help Group-Bank Linkage Programme, begun by NABARD in 1992 to link self-help groups to bank credit.

Q.17 · Answer (D) Swami Dayanada Saraswati
Swami Dayanand Saraswati, who founded the Arya Samaj in 1875, held the Vedas to be the source of all true knowledge and called for a return to them.

Q.18 · Answer (A) 65 years
Under Article 124(2) a Supreme Court judge holds office until the age of 65; High Court judges retire at 62.

Q.19 · Answer (A) Indo European language
The Rigveda, composed in early Vedic Sanskrit, is among the oldest texts in any Indo-European language.

Q.20 · Answer (A) Gurjar-Pratihara, Rashtrakuta and Palas
The Gurjara-Pratiharas, the Rashtrakutas and the Palas fought for control of Kannauj from the 8th to the 10th century.

Q.21 · Answer (D) Shiva
Nataraja, 'lord of dance', is Shiva in his form as the cosmic dancer.

Q.22 · Answer (C) Meenakshi
Madurai's Chithirai festival celebrates the marriage of the goddess Meenakshi to Sundareswarar (Shiva).

Q.23 · Answer (C) Carbon dioxide
The acetic acid in vinegar reacts with sodium bicarbonate to release carbon dioxide, which fills the balloon.

Q.24 · Answer (C) 2 R
All four in series give 4R and all in parallel 0.25R; two in parallel plus two in series give 0.5R + 2R = 2.5R, and three in parallel plus one in series R/3 + R ≈ 1.33R. The nine values possible with all four are 4R, 2.5R, 1.67R, 1.33R, R, 0.75R, 0.6R, 0.4R and 0.25R, so 2R cannot be made.

Q.25 · Answer (C) Chromosomes
Chromosomes, threads of DNA and protein in the nucleus, carry the genes and are seen as rod-shaped bodies only when the cell divides.

English Language and Comprehension

Q.1 · Answer (C) on
'Interested' takes 'in': 'They were interested in learning new languages.'

Q.2 · Answer (A) fell through due to lack
To 'fall through' is to fail to happen: 'The project fell through due to lack of funding and support.'

Q.3 · Answer (C) will singing
'Will' is followed by the base form of the verb: 'Sarah will sing at the concert on Saturday.'

Q.4 · Answer (B) complete the journey in mortification
After 'had to' both verbs take the base form, and 'in' needs a noun: 'had to swallow his bombast and complete the journey in mortification'.

Q.5 · Answer (D) Unwilling
'Reluctant' means unwilling or hesitant.

Q.6 · Answer (C) praise
'Kudos' is praise for an achievement.

Q.7 · Answer (D) Colloquiel
The correct spelling is 'colloquial'.

Q.8 · Answer (A) at
Something happens 'at' a meeting: 'impressed everyone at the meeting'.

Q.9 · Answer (A) was at your liking.
The idiom is 'to one's liking': 'I hope that yesterday's lunch at Rahul's place was to your liking.'

Q.11 · Answer (C) ropes
To 'learn the ropes' is to learn how a job is done or how things work.

Q.12 · Answer (D) was watched
The action was in progress, so the past continuous active is needed: 'While she was watching the movie, her phone rang.'

Q.13 · Answer (A) Hinder
To 'hamper' is to 'hinder', to get in the way of something.

Q.14 · Answer (A) Remedy
A 'panacea' is a remedy for all ills: a cure-all.

Q.15 · Answer (D) not as easy as it seems to be
'Easier said than done' means harder than it sounds: not as easy as it seems.

Q.16 · Answer (B) will be reshaping
An action going on at a future time takes the future continuous, 'will be reshaping'; technology does the reshaping, so the passive forms do not fit, and the present perfect 'have reshaped' does not suit a future year.

Q.17 · Answer (B) maintenance
The correct spelling is 'maintenance'.

Q.18 · Answer (C) composure
A possessive ('the employer's') must be followed by a noun: 'composure', calmness.

Q.19 · Answer (A) easygoing
'Lenient' means tolerant and not strict: 'easygoing'.

Q.20 · Answer (A) succeeded for all
The superlative of the adjective is needed, with 'of' for the group: 'the most successful of all the industrialists'.

Q.21 · Answer (D) a
'Fear is a natural response to danger': the indefinite article before a singular countable noun.

Q.22 · Answer (D) in
'In response to' is the fixed phrase: phobias often occur in response to something unlikely to cause real harm.

Q.23 · Answer (C) phobia
The passage is about phobias: 'a person with a phobia' may meet the criteria for a specific phobia.

Q.24 · Answer (B) experience
'Will' needs a verb, and one feels or 'experiences' fear: 'will experience intense fear and anxiety'.

Q.25 · Answer (C) characterised
Phobias 'are characterised by', that is marked by, significant distress.

This paper online, with tap-to-reveal answers: https://parikshasaathi.com/exams/ssc-mts/question-papers/mts-2024-8-oct-2024-shift-2/
Every solved SSC MTS paper: https://parikshasaathi.com/exams/ssc-mts/question-papers/
Daily current affairs PDF, free, every morning on Telegram: t.me/parikshasaathiapp
Questions and official answers: the Staff Selection Commission (SSC). Explanations: drafted with AI assistance by Pariksha Saathi and checked against the official answers. Pariksha Saathi is not affiliated with any exam body. Spotted an error? Write to sriwastava2@gmail.com.